

Surrogate-Accelerated Conjunction Assessment

Space-filling designs and Gaussian-process emulation for collision probability at catalog scale

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Abstract

Screening the tracked-object catalog for collision risk requires evaluating collision probability P_c for every close approach, and the standard Monte Carlo estimator is prohibitively expensive for the rare-event probabilities involved. We show that it is worse than expensive: across the realistic parameter range, P_c spans roughly thirteen orders of magnitude, and at 10^8 samples brute force returns *zero hits* – no estimate at all – for about three quarters of encounters. We build a Gaussian-process surrogate over the encounter-parameter space, trained at points chosen by parallel-tempering space-filling designs. Two exact invariances of the collision integral reduce its natural six-parameter description to four, a reduction verified numerically to $\sim 10^{-12}$ and worth an 8 to 14-fold saving in expensive evaluations. The surrogate attains 0.03 orders of magnitude RMSE in $\log_{10} P_c$ from 64 training runs and answers subsequent queries in 8 μ s, breaking even against Monte Carlo after 64 conjunctions and running ~ 5300 times cheaper over a 337 789-pair catalog screen. We report two negative results: space-filling designs cannot be substituted for Monte Carlo samples in the rare-event integral itself, and the 2–3 $\times$ advantage optimised designs hold on their own criterion does not translate into a proportional accuracy advantage.

1 Introduction

Two tracked objects pass close to one another. The probability that they collide is not a geometric question, because neither position is known precisely. Orbit uncertainty derived from two-line element sets is of order 1 km, while the objects themselves are of order 1 m and the reported miss distances are hundreds of metres. The uncertainty is three orders of magnitude larger than the objects and larger than the separation being resolved, so only a probability is meaningful.

We measured this gap directly. Rebuilding 37 published conjunctions by propagating both objects from public element sets to the time of closest approach gave separations differing from the reported miss distance by 0.6 km to 2.0 km, against reported values of 0.08 km to 0.43 km (Section 2). Every rebuilt separation was *larger* than reported, which is the expected consequence of adding isotropic error to a near-zero displacement: distance is a norm, so random perturbation almost always increases it.

Catalog-scale screening compounds the problem. Megaconstellation growth is increasing the number of screened pairs faster than compute budgets grow, and the standard Monte Carlo estimator scales as $1/\sqrt{N}$, so a tenfold accuracy improvement costs a hundredfold in samples.

Contributions.

1. A measurement of *where brute force fails outright* rather than merely slowly (Section 3, Figure 1).
2. An exact $6 \rightarrow 4$ reduction of the encounter-parameter space, verified numerically rather than asserted (Section 5).
3. A Gaussian-process surrogate trained at parallel-tempering space-filling design points, benchmarked against random Latin hypercubes and uniform sampling (Section 6).
4. A triage classifier trained on a catalog screen of 337 789 conjunctions, evaluated at full recall against the baseline an analyst would actually use (Section 7).
5. Two negative results (Sections 4 and 6) that constrain how this family of methods should be applied.

2 Propagation and uncertainty

Element sets are retrieved from CelesTrak and Space-Track and propagated with SGP4 [1, 2]. TLE elements are Brouwer mean elements defined by the SGP4 theory itself; propagating them with a general-purpose orbit integrator introduces kilometre-level error immediately, so the element set and the propagator are used strictly as a matched pair. Propagation was validated against the International Space Station: mean altitude 419.4 km and period 92.96 min from the elements, against 92.91 min measured independently by counting minima of geocentric radius over three days of propagation.

The public conjunction feed carries no covariance data. It reports time of closest approach, miss distance, object identifiers, radar cross-section class and the originator’s own probability, but no state vectors and no covariance matrices. Uncertainty must therefore be modelled, and we state it as an assumption rather than presenting it as a data product.

Each object is assigned a diagonal covariance in its own radial / in-track / cross-track (RTN) frame,

$$C_{\text{RTN}} = \text{diag}(\sigma_R^2, (k_T \sigma_R)^2, (k_N \sigma_R)^2), \quad (1)$$

with anisotropy ratios fixed at $k_T = 10$, $k_N = 1.5$ and only the scale σ_R fitted. In-track error dominates because a small period error integrates into a growing along-track lag. Fixing the ratios is deliberate: a full 3×3 covariance per object has twelve free parameters, and the data provides one scalar observable per event. Fitting the scale alone to the rebuilt-versus-reported miss spread gives

$$\sigma_R = 95 \text{ m}, \quad \sigma_T = 953 \text{ m}, \quad \sigma_N = 143 \text{ m},$$

magnitudes consistent with published TLE accuracy and obtained from our own measurements rather than assumed. The two objects occupy different orbital planes, so each covariance is rotated into a common inertial frame before the relative covariance $C = C_1 + C_2$ is formed.

What this model does not do. Per-event correlation between predicted and observed error is -0.53 . The model reproduces the aggregate *scale* of TLE uncertainty but not the per-event ranking; it does not identify which conjunctions are worst. A single scalar observable per event is insufficient to constrain that, and we do not claim otherwise.

3 Collision probability and the cost of brute force

At the relative speeds typical of low Earth orbit (10 km s^{-1} to 15 km s^{-1}) an encounter lasts milliseconds and relative motion is effectively rectilinear. The three-dimensional problem then collapses onto the two-dimensional encounter plane perpendicular to the relative velocity [3, 4], and

$$P_c = \int_{\|x\| \leq R} \mathcal{N}(x; \mu, C) dx, \quad (2)$$

where μ and C are the projected relative miss vector and covariance and R is the combined hard-body radius.

We implement three independent estimators of (2): a batched Monte Carlo sampler, polar quadrature with resolution adapted to the covariance, and a small-disk closed form valid when $R \ll \sigma$ [5]. Monte Carlo and quadrature agree within Monte Carlo error bars on every real conjunction tested (maximum $|z| = 1.4$), which cross-validates the encounter geometry and the sampler against each other.

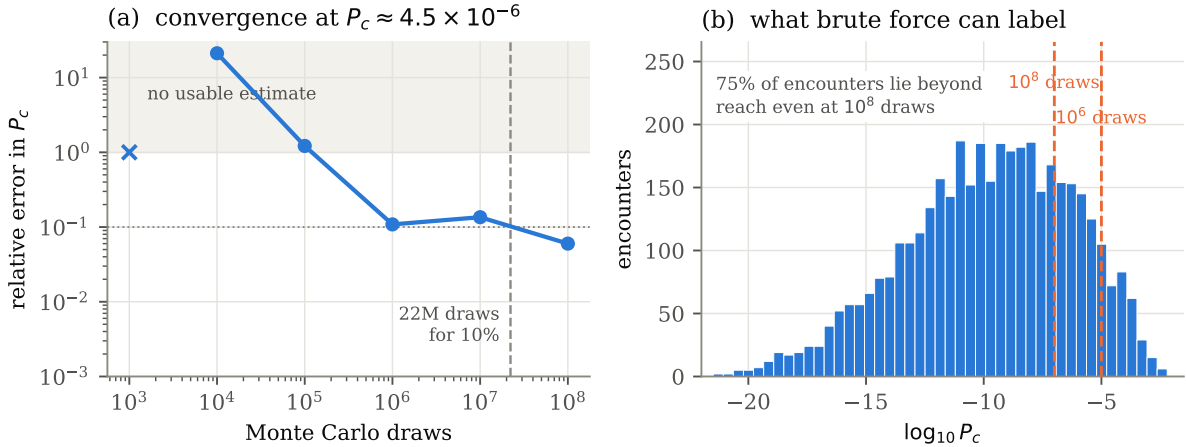


Figure 1: Brute force does not merely converge slowly; over most of the parameter range it returns nothing. **(a)** Monte Carlo convergence at a representative $P_c \approx 4.5 \times 10^{-6}$: below 10^5 samples the estimator produces zero hits, and 22 000 000 samples are needed for 10 % relative error. **(b)** Distribution of P_c over the encounter-parameter box, with the reach of two sample budgets marked at ten expected hits. Roughly three quarters of encounters lie beyond reach even at 10^8 samples.

Figure 1 states the problem quantitatively. At $P_c \approx 4.5 \times 10^{-6}$, 100 000 samples return *zero hits*: not a poor estimate but no estimate. Reaching 10 % relative error requires $\sim 22\,000\,000$ samples and 1 % requires $\sim 2.2 \times 10^9$, per conjunction. Across the parameter box P_c spans about thirteen orders of magnitude, and a substantial fraction of encounters sit below 10^{-12} , where no achievable sample budget yields a single hit.

This has a direct methodological consequence: **training labels for the surrogate cannot come from Monte Carlo**, because Monte Carlo cannot label most of the space. We use the quadrature, which is exact in this Gaussian setting and agrees with Monte Carlo wherever Monte Carlo works at all. The sampler stands in for the general case – non-Gaussian uncertainty, long-duration encounters, non-spherical bodies – where no closed form exists.

4 A negative result: designs cannot replace samples

The intuitive reading of “use a space-filling design instead of Monte Carlo” is to replace the random samples in (2) with a well-spread deterministic design. This does not work, and the reason is worth stating because the idea is natural.

Equation (2) is the integral of an indicator function over a microscopic region. A 256-point space-filling design returns zero hits just as 10^5 random samples do, and in fact more reliably, because such a design deliberately spreads points *away* from any small region. Quasi-Monte Carlo does not rescue the construction either: its convergence advantage requires a smooth integrand, and an indicator over a tiny region is the opposite of smooth.

The construction that does work places the design in the **encounter-parameter space** rather than the sample space. Each design point receives one expensive evaluation, a surrogate interpolates between them, and the cost is amortised across every subsequent query.

5 Dimensional reduction

The integral (2) appears to depend on six parameters: μ (two components), C (three), and R (one). Only four are independent. The integration domain $\|x\| \leq R$ is rotation-symmetric, so rotating μ and C together leaves P_c unchanged; and the integral is covariant under uniform scaling of all lengths, so dividing μ , $\sqrt{C}$ and R by a common factor leaves it unchanged. Rotating into the eigenbasis of C and scaling by R therefore removes two parameters, leaving $(\mu_1/R, \mu_2/R, \sigma_1/R, \sigma_2/R)$.

Because a silently incorrect reduction would invalidate everything downstream, the invariances were verified numerically rather than accepted from the algebra (Table 1).

Table 1: Numerical verification of the reduction over random encounters spanning the full parameter box. Residuals are at the level of quadrature noise: these are exact identities, not approximations.

Claim	Cases	Max. relative error
P_c invariant under rotation of μ and C	300	8.3×10^{-13}
P_c invariant under uniform length scaling	300	9.8×10^{-13}
Reduced form reproduces P_c	200	8.4×10^{-13}
P_c even in each μ component (eigenbasis)	200	6.2×10^{-16}
$6 \rightarrow 4 \rightarrow P_c$ round trip	200	5.5×10^{-13}

The saving is substantial. Measured fill distance at $k = 4$, $n = 64$ is 0.397; matching it in six dimensions requires ~ 500 to 900 points, an 8 to 14-fold increase in expensive evaluations for identical answers. The sign symmetry, exact to 6×10^{-16} , permits restricting the domain to $\mu_1, \mu_2 \geq 0$, reducing its volume by a further factor of four.

The identity holds *given the model*: a circular hard body, Gaussian uncertainty, and the short-term encounter approximation. Slow encounters, below about 1 km s^{-1} relative speed, fall outside it and are excluded here.

6 Surrogate

Design points are generated by parallel-tempering [9] optimisation of the MaxPro criterion [8], reusing a C++ implementation developed for unrelated prior work and wrapped for Python. Designs are static artefacts: the optimiser runs once per (n, k) and costs 0.2s to 1.4s. On its own criterion the optimised design beats the best of twenty random Latin hypercubes [6] by 2.1–3.1 $\times$; the Morris–Mitchell ϕ_p criterion [7] is reported alongside it in the project log.

The surrogate is a Gaussian process [10] with an anisotropic squared-exponential kernel, hyperparameters obtained by maximising the log marginal likelihood. Targets are $\log_{10} P_c$: with thirteen orders of magnitude of dynamic range, a GP fitted to P_c itself would be dominated entirely by the few largest values.

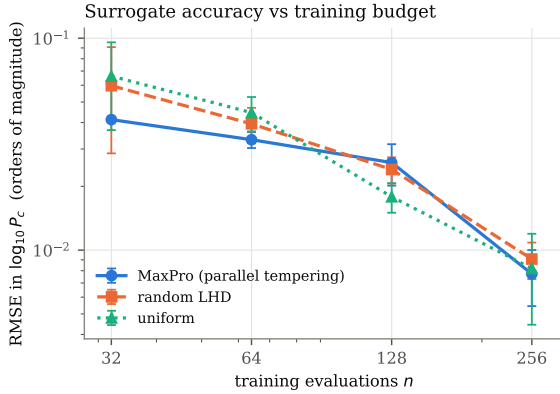


Figure 2: Surrogate accuracy against training budget, five replicates per design type; error bars are the standard deviation across replicates. All three converge by $n = 256$.

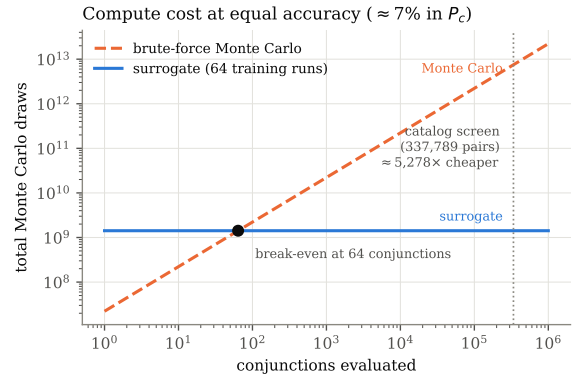


Figure 3: Total compute to evaluate Q conjunctions at matched accuracy ($\approx 7\%$ in P_c). The surrogate’s cost is flat after training; break-even is 64 conjunctions.

At $n = 64$ the surrogate attains an RMSE of about 0.03 orders of magnitude in $\log_{10} P_c$ – roughly 7% in P_c – and answers subsequent queries in 8.3 μ s. Figure 3 converts this into the quantity that matters at catalog scale. Charging the surrogate its full training cost and Monte Carlo its per-query cost at matched accuracy, the two break even at 64 conjunctions; over the 337 789-pair screen of Section 7 the surrogate is ~ 5300 times cheaper.

A second negative result. The 2–3 $\times$ advantage the optimised designs hold on the MaxPro criterion *does not* translate into a proportional accuracy advantage. Mean RMSE improves by only 1.1–1.8 $\times$ over random Latin hypercubes and uniform sampling at $n \leq 128$, and by $n = 256$ the three are indistinguishable (Figure 2). The target function is smooth, so once the box is adequately covered the design ceases to matter. Better space-filling by the criterion is not the same thing as better surrogate accuracy.

Where the optimised designs do show a consistent advantage is **variance**. Across replicates their RMSE spread is several times tighter than that of random designs. With a one-shot expensive evaluation budget a random design is a lottery – some draws are as good as an optimised one, others substantially worse – and reproducibility is worth as much to a practitioner as the mean improvement.

7 Risk triage

The public conjunction feed contains no negative examples: every record is flagged reportable with $P_c > 10^{-4}$. A classifier trained on it learns to answer “yes”. We therefore manufactured negatives by screening debris catalogs directly – dense, co-orbital populations that generate many close approaches per object.

Screening uses a coarse propagation grid, per-pair local minima of separation detected on a rolling three-step window, an analytic linear-motion filter, and parabolic refinement of the survivors. The coarse gate must be wide: at 16 km s^{-1} a 60 s grid steps 960 km, so a pair passing within 1 km may never appear closer than $\sim 450 \text{ km}$ at any sampled instant. The gate is consequently set to 530 km, which leaves nearly every pair a candidate; the analytic filter, which locates the rectilinear closest approach in closed form, then removes 91 % of candidates with identical output.

The resulting dataset holds 337 789 conjunctions from three debris families over seven days, with 352 positives (0.104 %) at $\log_{10} P_c > -10$. No conjunction in this population reached the operational 10^{-4} threshold, so the label threshold is a stated choice; the ordering of methods is unchanged at 10^{-8} and 10^{-12} .

Features are restricted to quantities available *before* any P_c computation – miss distance, relative speed, altitude, relative inclination, element-set age, period difference, eccentricity and drag term. Including P_c or anything derived from it would make the task circular.

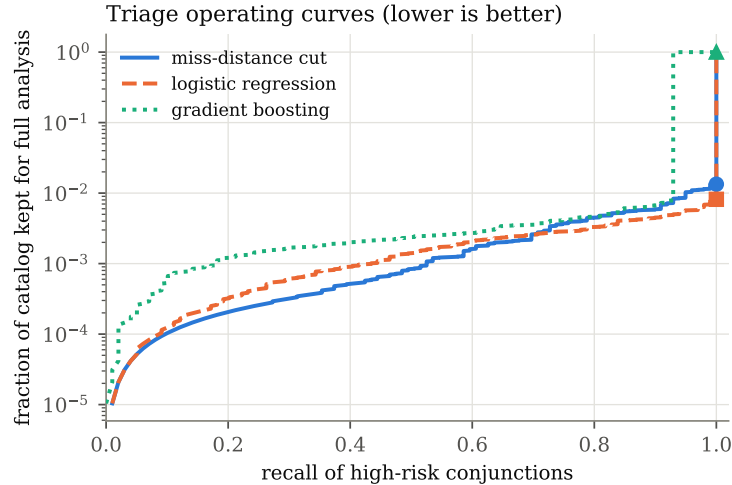


Figure 4: Triage operating curves on held-out data. Below 70 % recall a miss-distance cut is the better ranker; the learned model wins at the high-recall end, which is the operationally relevant one. Gradient boosting reaches full recall only by retaining the entire catalog.

Accuracy and AUC are near-meaningless at a 0.1 % positive rate; a constant negative prediction scores 99.9 %. Because missing a high-risk conjunction is unacceptable while passing a harmless one costs only compute, models are compared by how much of the catalog they discard *at full recall*. The baseline is a miss-distance cut, which is what an analyst does without a model, and splits are grouped by day or by object rather than at random, since conjunctions in the same window share objects and geometry.

Logistic regression passes 0.82 % to 0.92 % of conjunctions against 1.15 % to 1.41 % for the baseline, and on the object split it retains full recall where the baseline misses one. Miss distance

Table 2: Triage performance. “Kept” is the fraction of conjunctions passed on for expensive analysis; lower is better at equal recall. Thresholds are set from out-of-fold predictions on training data and applied unchanged.

Split	Model	Kept	Recall	Missed
Day	Miss-distance cut (baseline)	1.41 %	100.0 %	0
	Logistic regression	0.92 %	100.0 %	0
	Gradient boosting	100.0 %	100.0 %	0
Object	Miss-distance cut (baseline)	1.15 %	97.2 %	1
	Logistic regression	0.82 %	100.0 %	0
	Gradient boosting	100.0 %	100.0 %	0

alone has a rank correlation of only +0.60 with $\log_{10} P_c$: the projected covariance depends on encounter geometry that a single distance does not capture.

Gradient boosting fails outright, retaining the entire catalog under every configuration. This is not a threshold artefact – its oracle operating point is also $\approx 100\%$, meaning the lowest-scoring true positive ranks below almost the whole test set. At a 0.1 % positive rate the flexible model fragments a space in which the positives occupy a thin tail and cannot rank them, while a model whose score is monotone in roughly the right direction can. Model capacity actively hurts.

8 Limitations

1. **Absolute probabilities are not validated.** Our P_c values run 10–100 $\times$ below the originator’s published figures. The dominant suspect is the hard-body radius, since $P_c \propto R^2$ and radar-cross-section-derived radii of 1 m to 4 m are likely below operational values. Agreement with published probabilities is not claimed as validation.
2. **Uncertainty is synthesized, not measured,** and reproduces the aggregate scale of TLE error rather than its per-event variation.
3. **The triage population is debris,** which is small, uncontrolled and co-orbital; operational payload conjunctions differ. Test sets contain 36–99 positives, so differences of a few tenths of a percent should not be over-read.
4. **Seven-day propagation from a single epoch** exceeds the 2 to 5-day range over which the covariance model was calibrated.
5. **The dimensional reduction is model-conditional** and does not extend to non-Gaussian uncertainty, non-circular bodies, or long-duration encounters.

9 Reproducibility

All results are produced by scripts in the accompanying repository. Every important quantity is computed two independent ways: Monte Carlo against quadrature, the C++ design criterion against a `numpy` reimplementation (agreeing to six decimal places), and the orbital period from elements against a period measured by counting radius minima. The test suite comprises 86 offline checks.

Table 3: Commands reproducing each result.

Command	Produces
<code>python src/validate_iss.py</code>	Propagation validation (Sec. 2)
<code>python src/calibrate.py</code>	Covariance calibration (Sec. 2)
<code>python src/run_montecarlo.py</code>	Cost baseline (Sec. 3)
<code>python tests/test_paramspace.py</code>	Table 1
<code>python src/run_surrogate.py</code>	Figure 2
<code>python src/build_dataset.py</code>	Triage dataset (Sec. 7)
<code>python src/run_triage.py</code>	Table 2
<code>python src/make_figures.py</code>	All figures

References

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